Accurate Kinetic Energy
No.11-3 Jianguo Rd., Tanzi Dist., 42760 Taiwan

All dimensions are millimeters.

CAD:	TCR
Review:	EG
Appr:	JL
Page: 1/4	Date: April 16,2024

Specification Title:

Clock Oscillator
LVDS - Complimentary Output
3.2 x 2.5 millimeter Surface Mount
General Product Specification

Part Number:	**S3A LVDS Series**	

Electrical Specifications:

Frequency Range		13.500 ~ 200.000				MHz
Frequency Stability		±25 ~ ±100				ppm
Aging per Year		±3				ppm Max.
Operating Temperature Range	Standard	-20 ~ +70				°C
	Option	-40 ~ +85				
	Option	-40 ~ +105				
	Option	-40 ~ +125				
Storage Temperature Range		-55 ~ +125				
Supply Voltage		1.8 ± 5%	2.5 ± 5%	3.3 ± 5%	2.375~3.63	VDD
Input Current		20	20	20	20	mA Max.
Output Voltage	Logic High (Voh)	1.60				VDD Max.
	Logic Low (Vol)	0.90				VDD Min.
Differential Output Voltage (V_{OD})		247 ~ 454				mV
Offset Voltage (V_{OS})		1.125 ~ 1.375				V
Output Symmetry (Duty Cycle)		45 ~ 55				%
Output Type		LVDS				
Output Load (Between Q and QN)		100				ohm
Rise and Fall Time	13.500 to 99.999MHz	1.0	1.0	1.0	1.0	ns Max.
	100.000 to 200.000MHz	0.5	0.5	0.5	0.5	
Enable-Disable Function		Tri-State				
Input Enable Voltage		1.26	1.75	2.31	70%VDD	VDD Min.
Input Disable Voltage		0.54	0.75	0.99	30%VDD	VDD Max.
Phase Jitter (12 KHz to 20 MHz)		0.5				ps Max. RMS

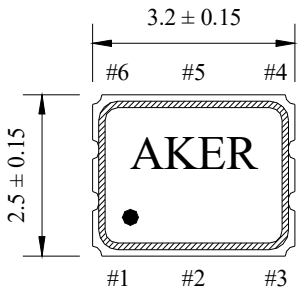
Temperature stability is Inclusive of all conditions:
Calibration Tolerance at +25°C, frequency stability over the operating temperature range, supply voltage change, output load change, shock, vilbration, and 1st year aging at +25°C.

RoHS Compliant
Pb - Lead Free

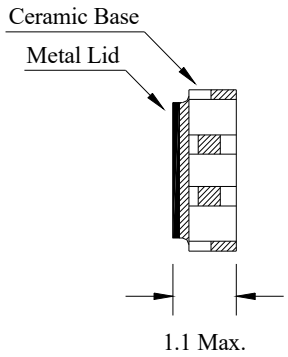
Ltr	Revisions	Date	Appr

Mechanical Outline and Solder Pad Layout:

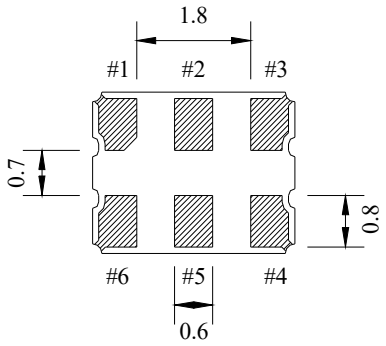
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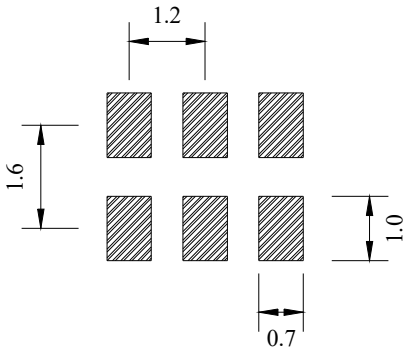
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<SUGGESTED LAYOUT>

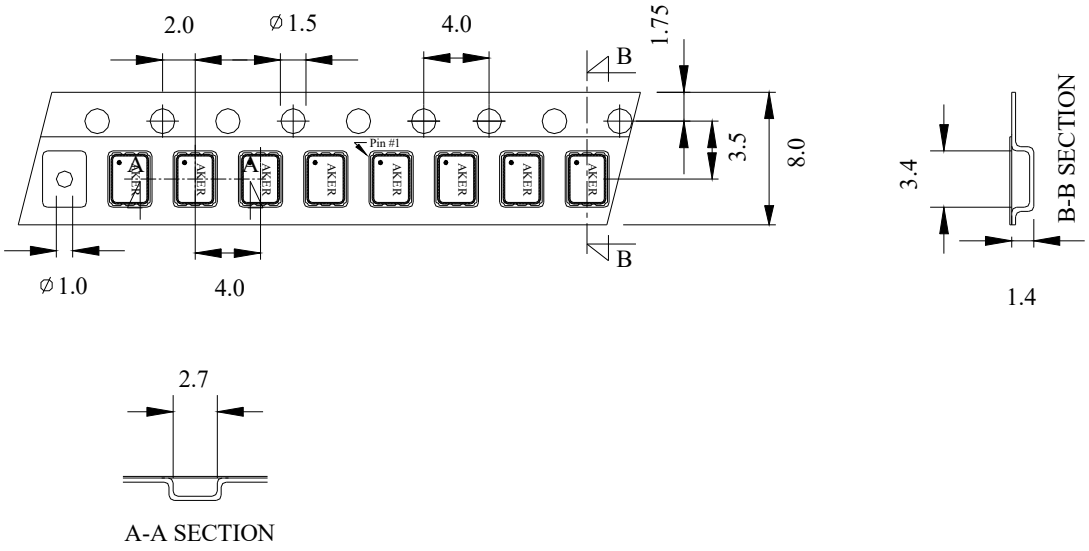


Pin Connection	
PIN No.	Connection
#1	Enable/Disable
#2	NC
#3	GND
#4	Output - Q
#5	Output - QN
#6	VDD

Enable/Disable Function	
PIN #1	PIN #4 & #5
HIGH or OPEN	Operating
LOW	High Impedance

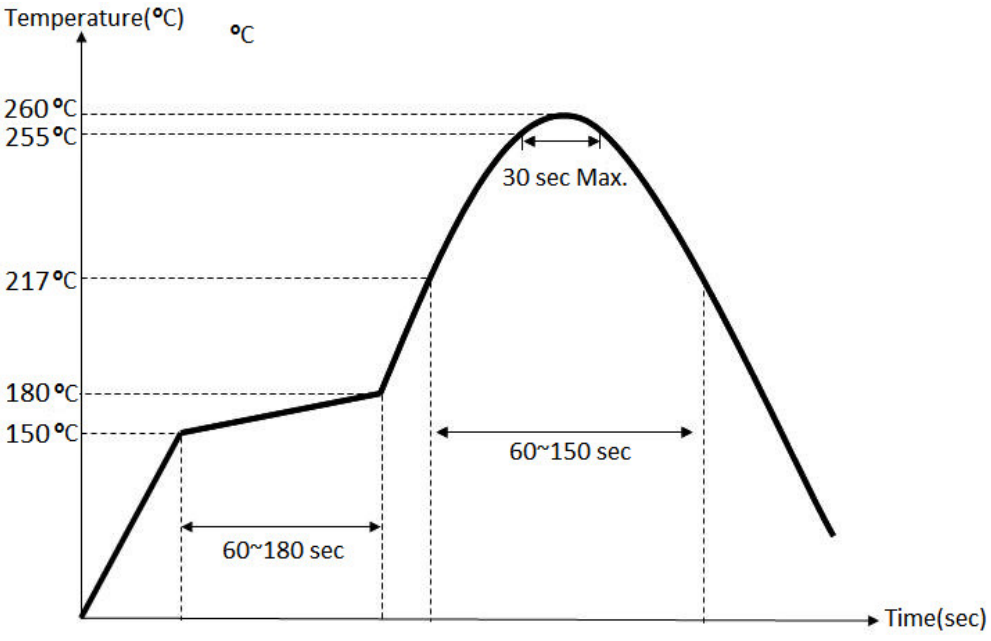
Package is Seam Sealed Ceramic-Metal.
Terminator Pads are Ni/Au.
Dimensions are millimeters.

Carrier Tape Dimensions:



Dimensions are millimeters.

Solder Reflow Characteristics:



How to build a Part Number:

Series	S	Parameter
Package	3A	3.2 x 2.5 mm - 6 Pad
Supply Voltage	33	+3.3 VDD ± 5%
	25	+2.5 VDD ± 5%
	18	+1.8 VDD ± 5%
	M	+2.375 ~ +3.63 VDD
Temperature Stability	10	±100 ppm
	05	±50 ppm
	025	±25 ppm
	-	
Frequency	13.500~200.000	MHz
	-	
Output Load	L	LVDS
	-	
Temperature Range	See Notes	-20 ~ +70 °C
	X	-40 ~ +85 °C
	X1	-40 ~ +125 °C
	X2	-40 ~ +105 °C
	-	
Packaging	M	250pcs Reel
	R	1000pcs Reel

Part Number Example:

S3A3305-156.250-L-X-R

S3A: 3.2 x 2.5 mm SMD Package - 6 Pad

33: +3.3±5% VDD Supply Voltage

05: ±50 ppm Temperature Stability

156.250 MHz Nominal Frequency

L: LVDS Output

X: -40 ~ + 85°C Temperature Range

R: Tape and Reel Packaging - 1000pcs Reel

Notes:

1- Standard Duty Cycle and Temperature Range do not need to be included in Part Number description.

2- Product is shipped in Tape and Reel configuration.

3- Quantities less than 250pcs are shipped in tape only.

4- Specification subject to change without notice.